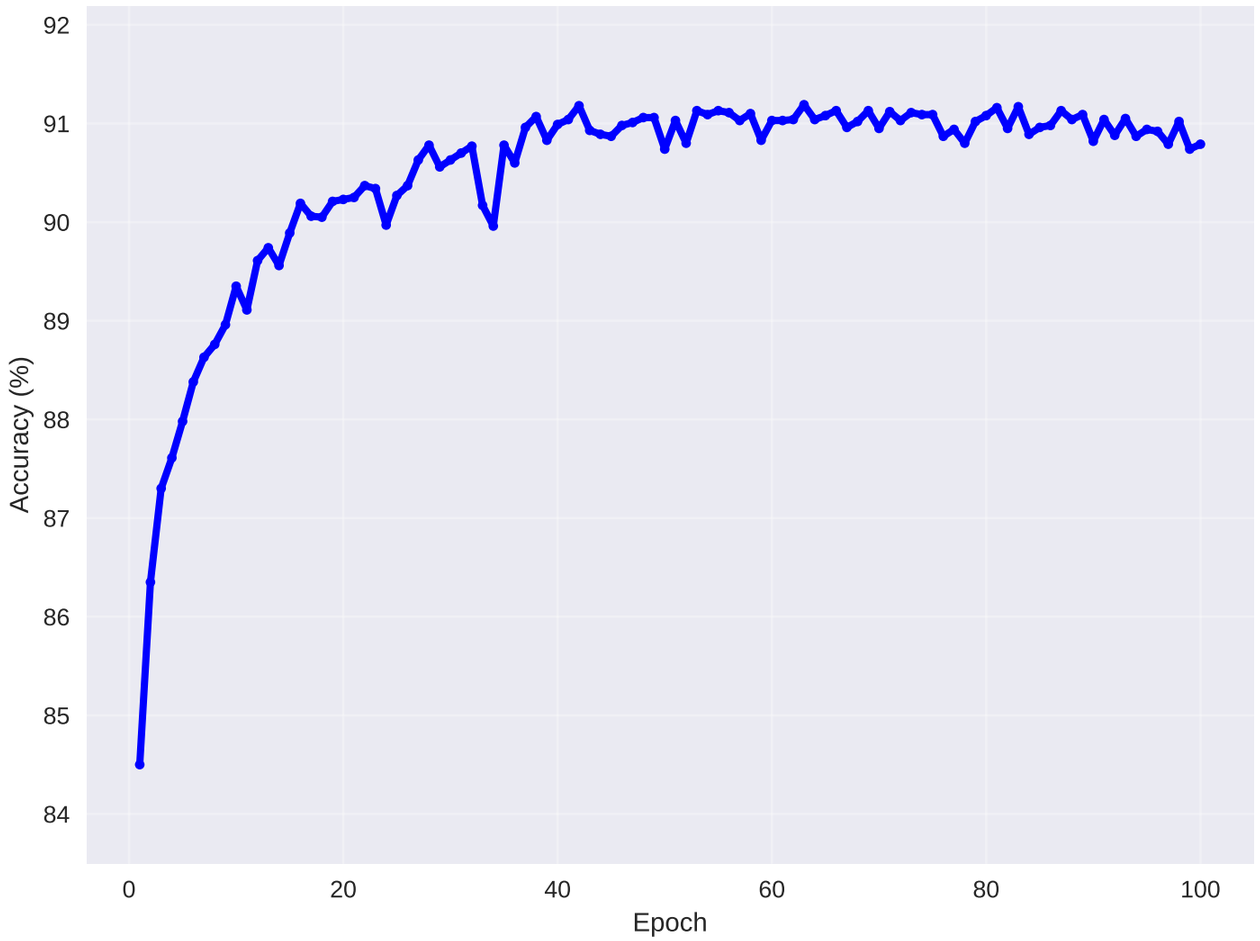
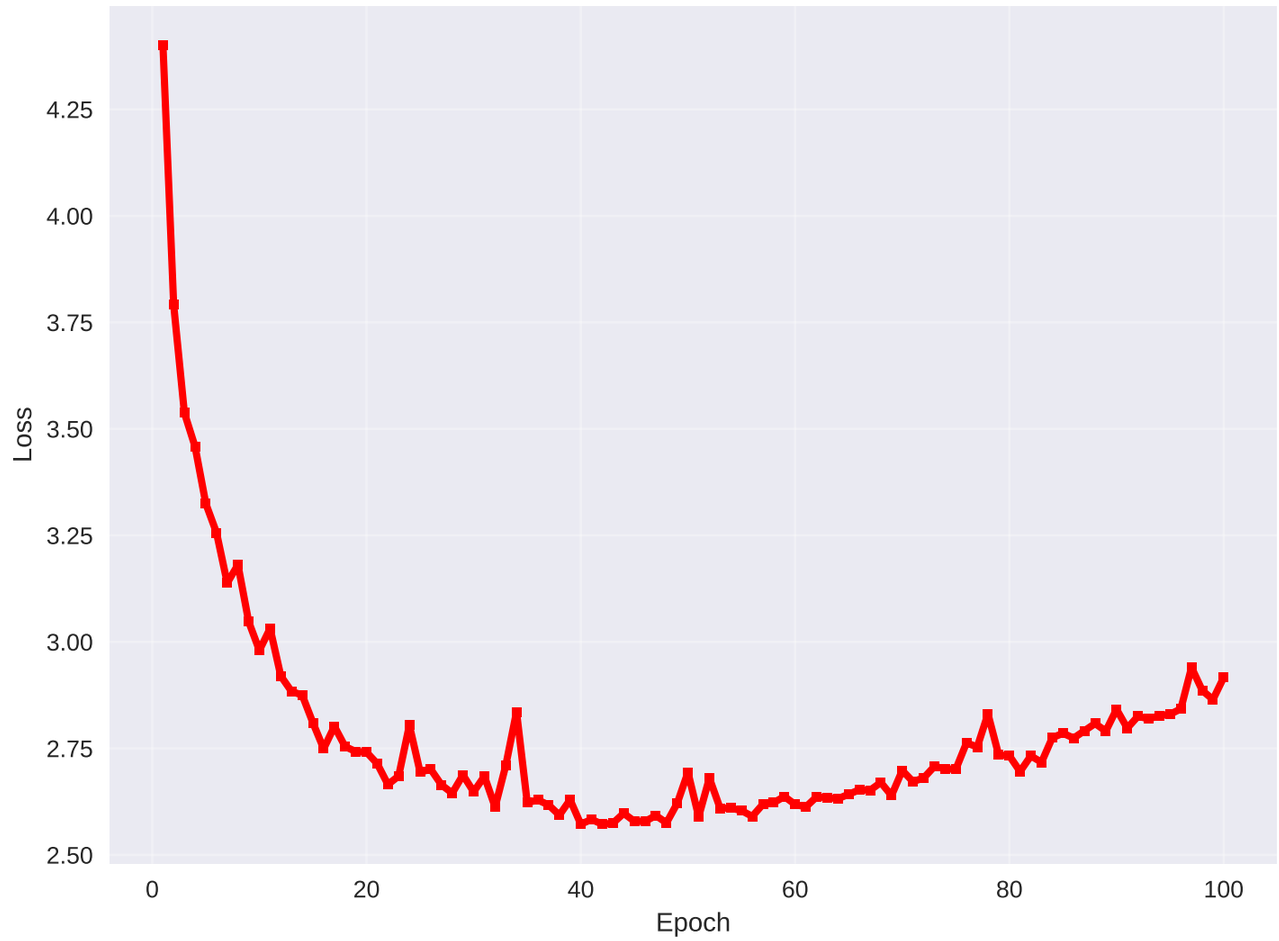
Federated Learning - Main Metrics Overview

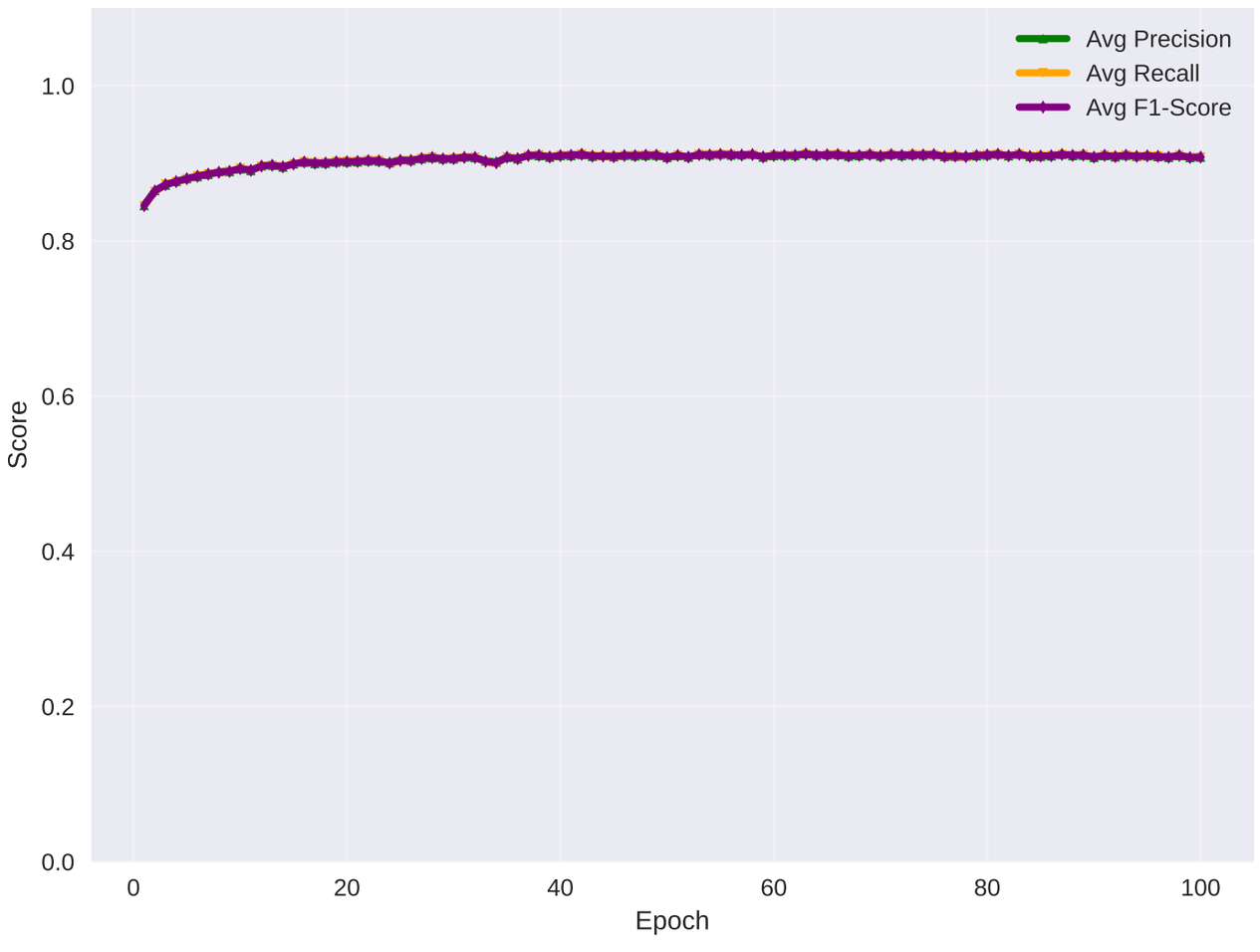
 Global Model Accuracy Evolution



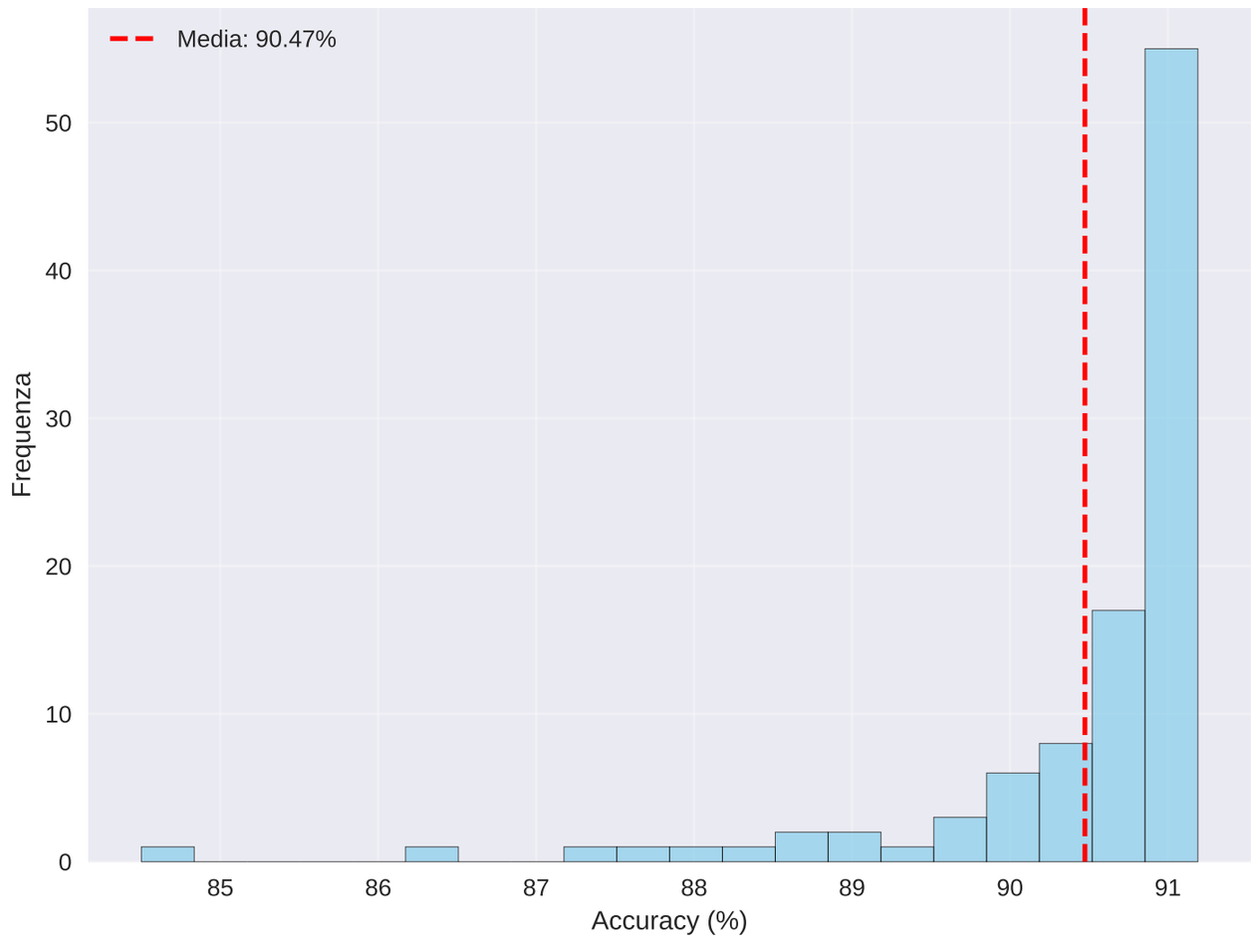
 Global Model Loss Evolution



 Average Precision, Recall & F1-Score

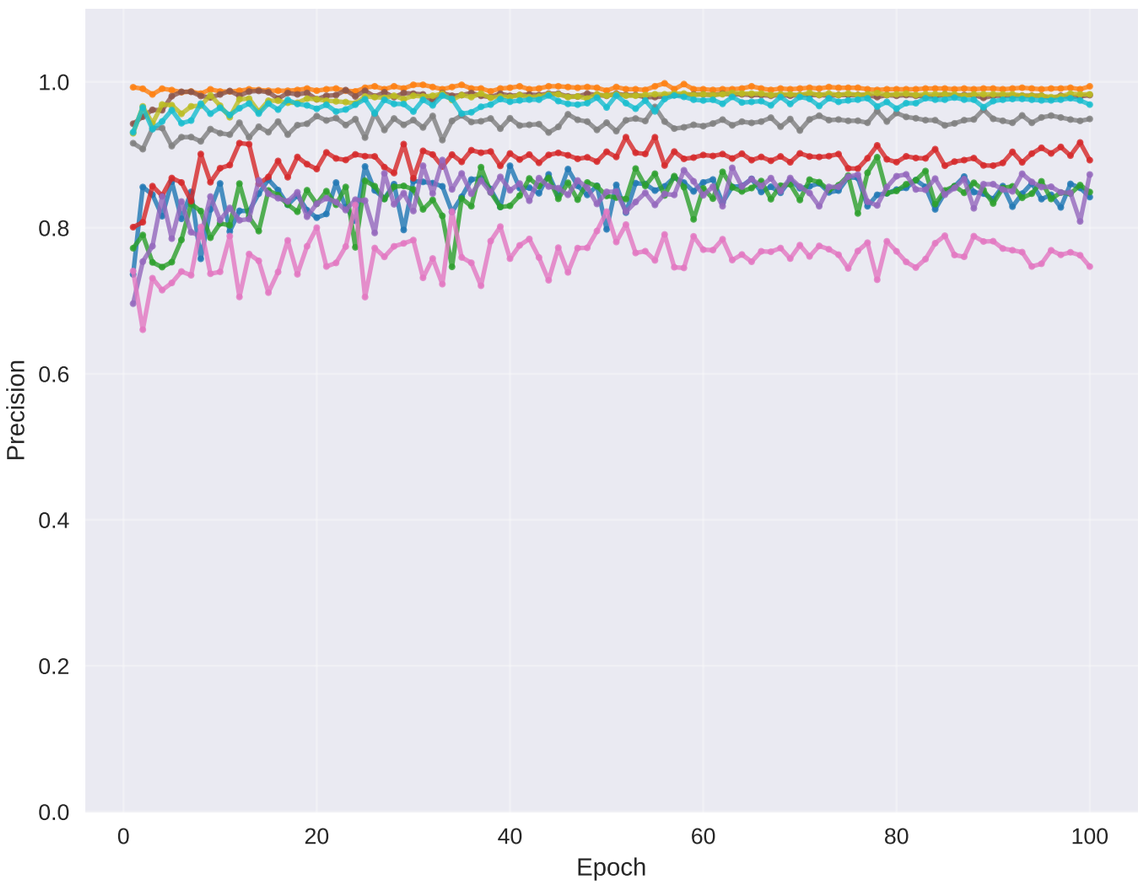


 Distribuzione Accuracy

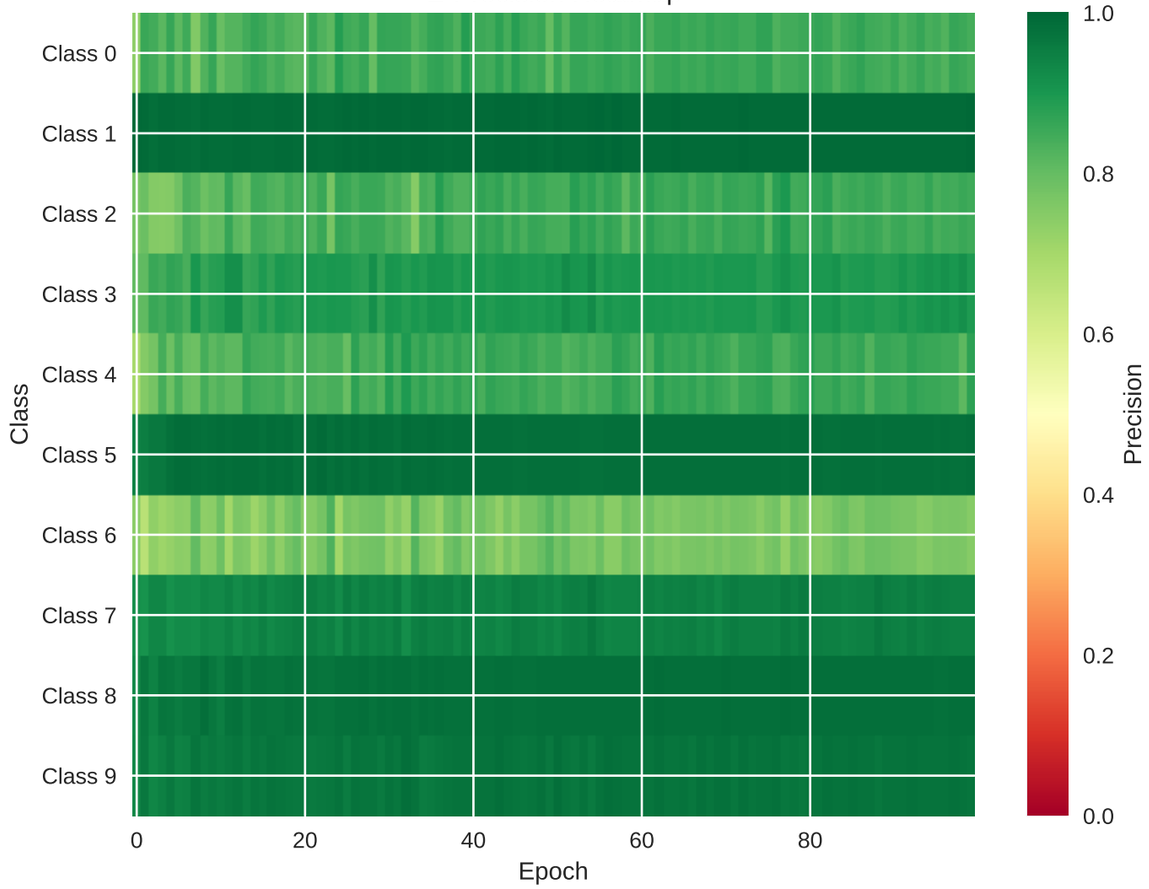


Per-Class Precision Analysis

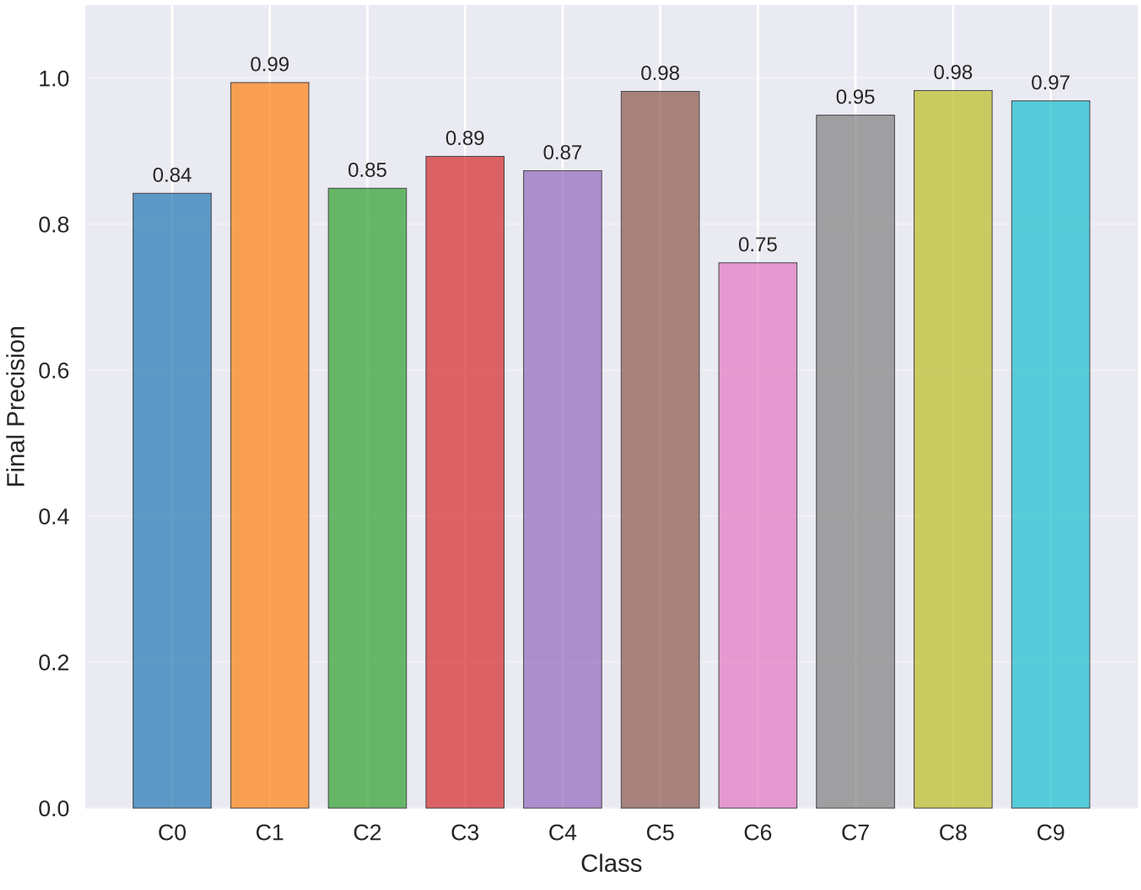
□ Per-Class Precision Evolution



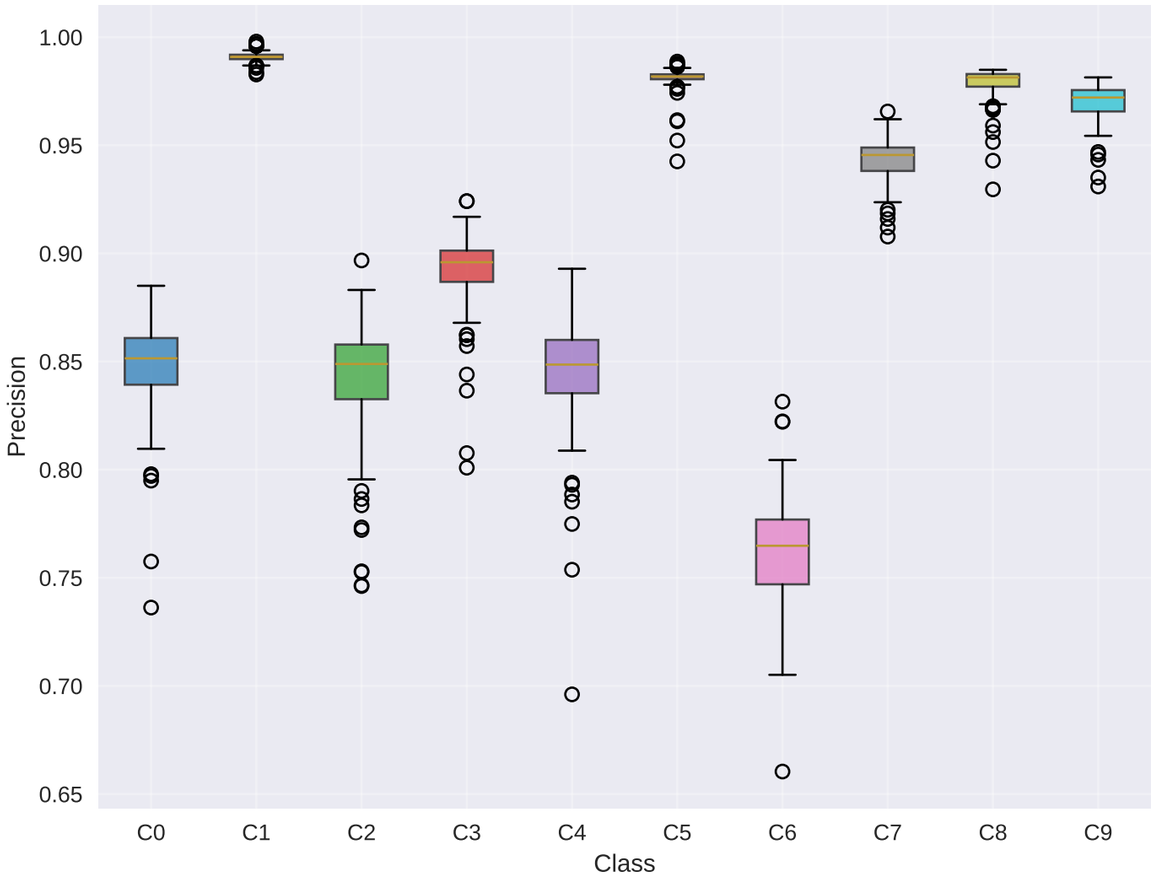
□ Precision Heatmap



□ Final Precision per Class

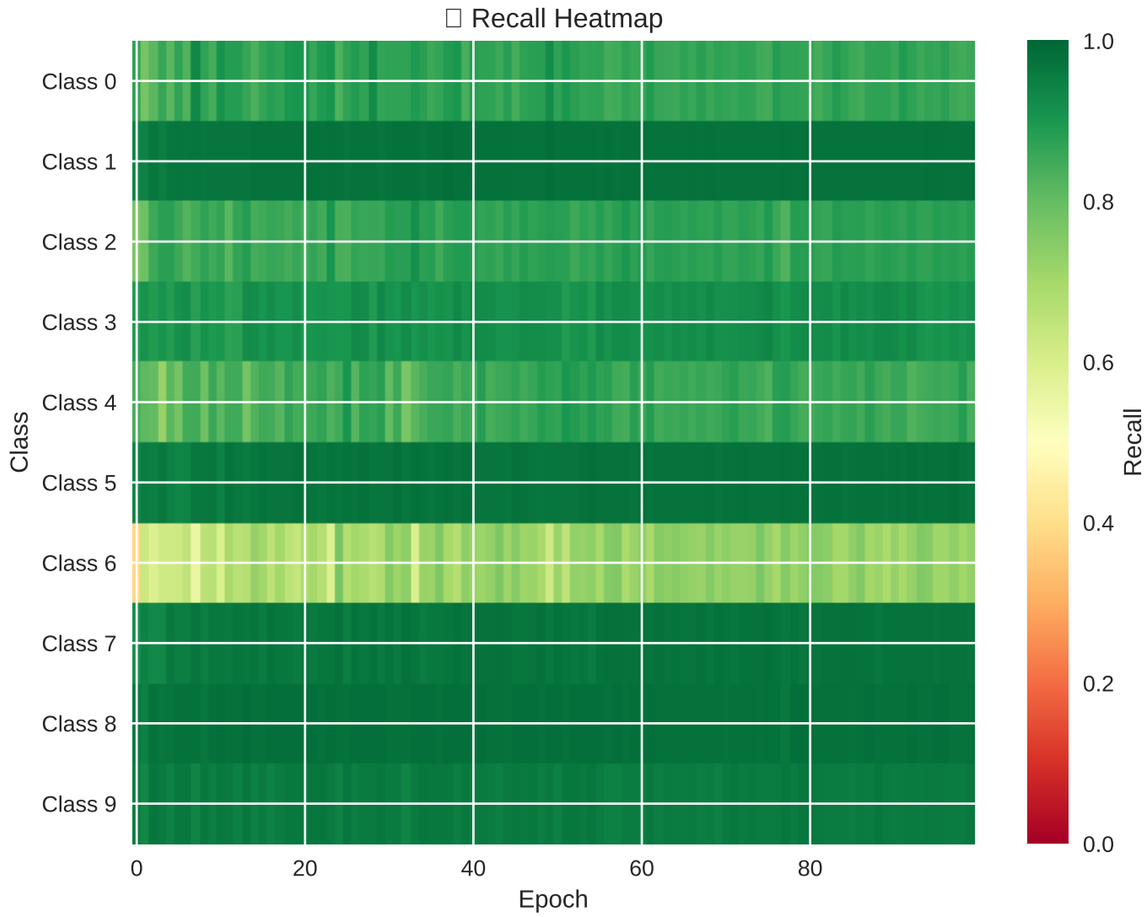


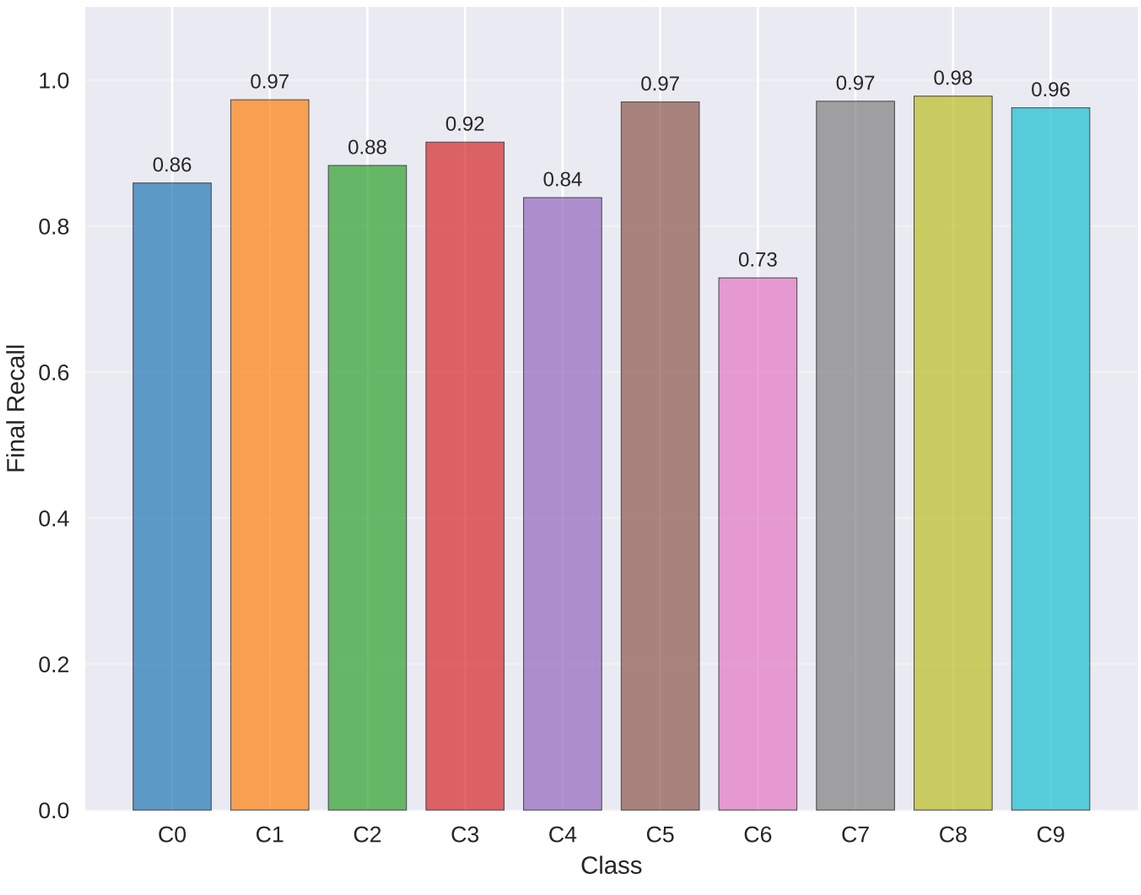
□ Precision Distribution per Class

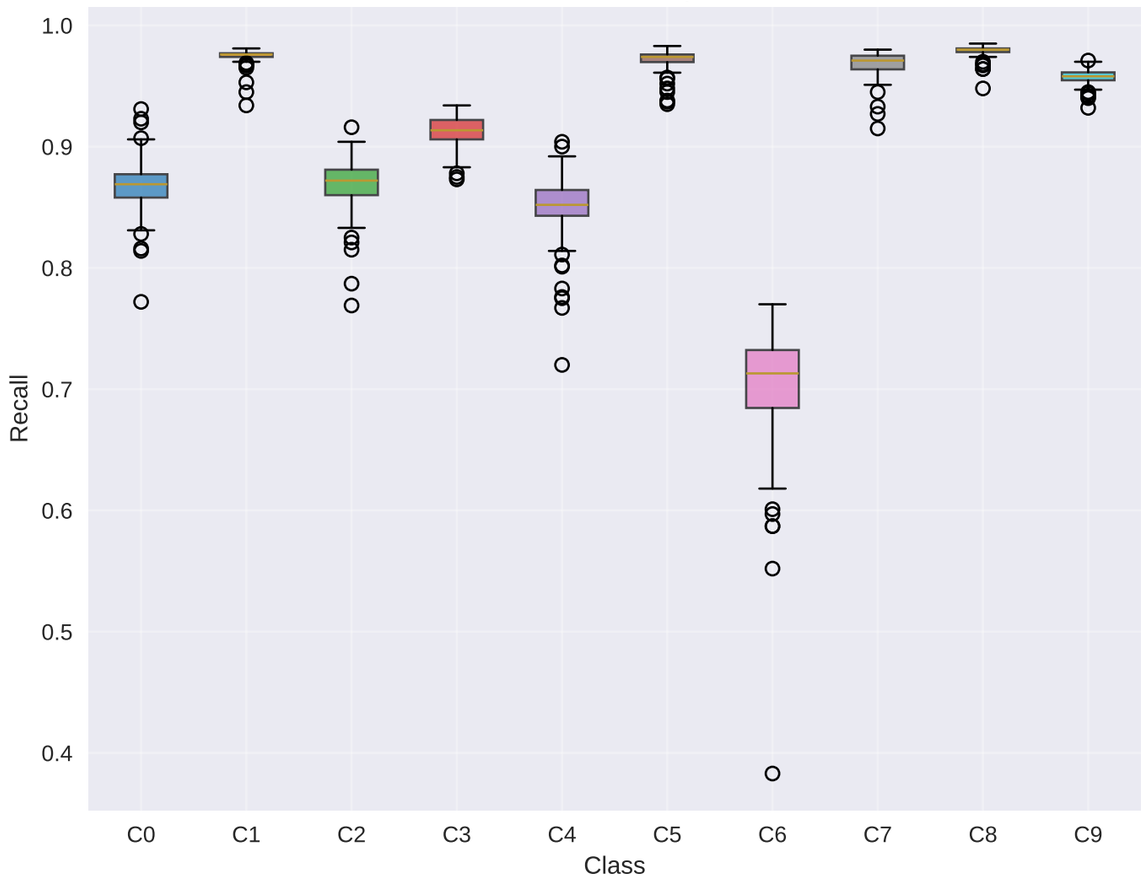


Per-Class Recall Analysis

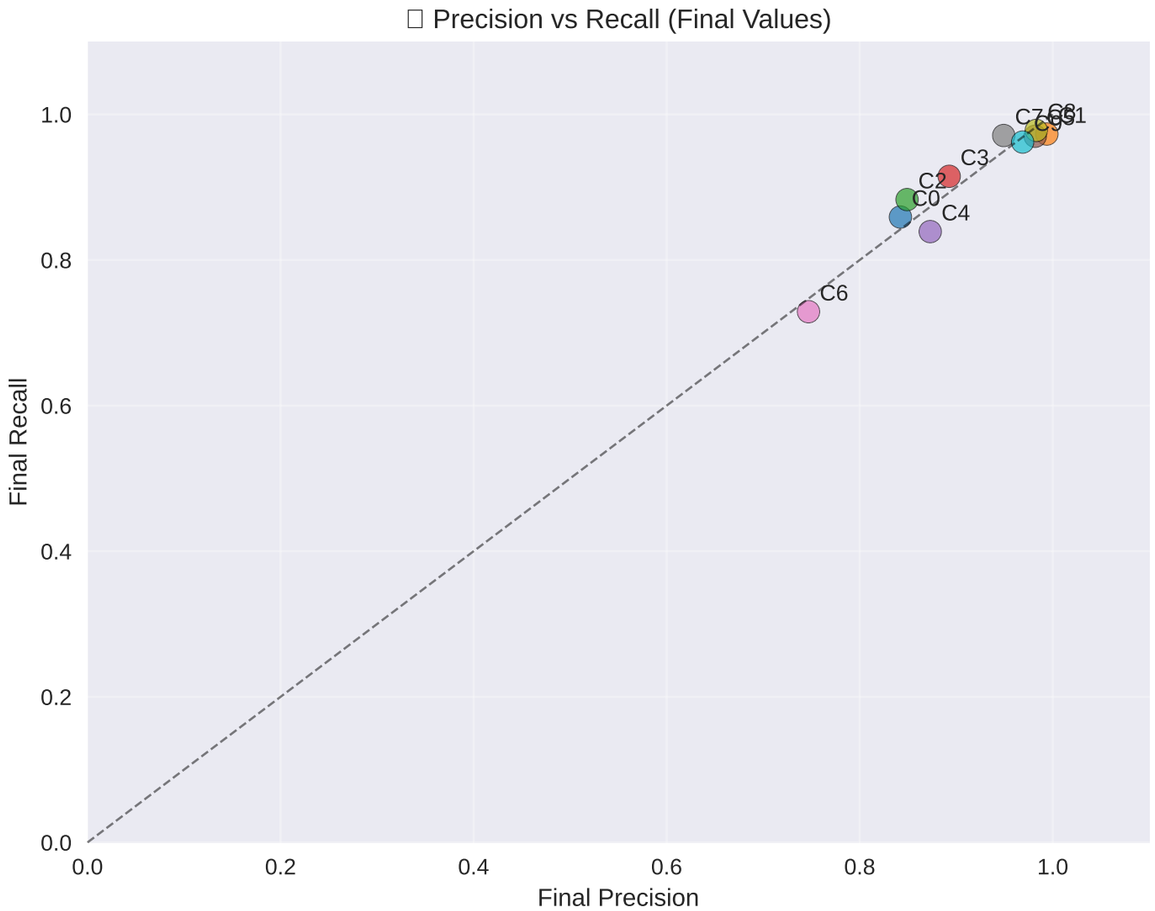


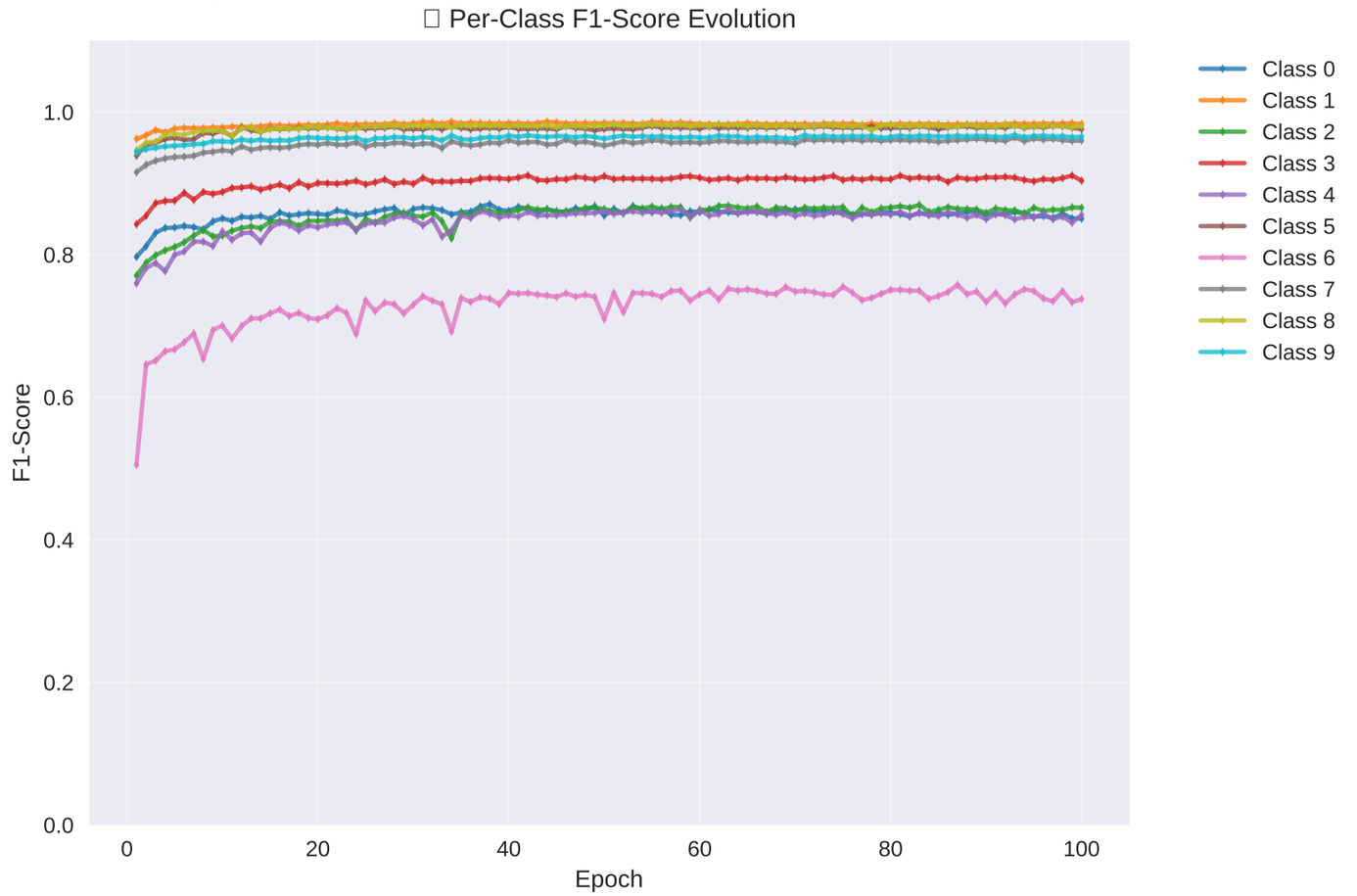


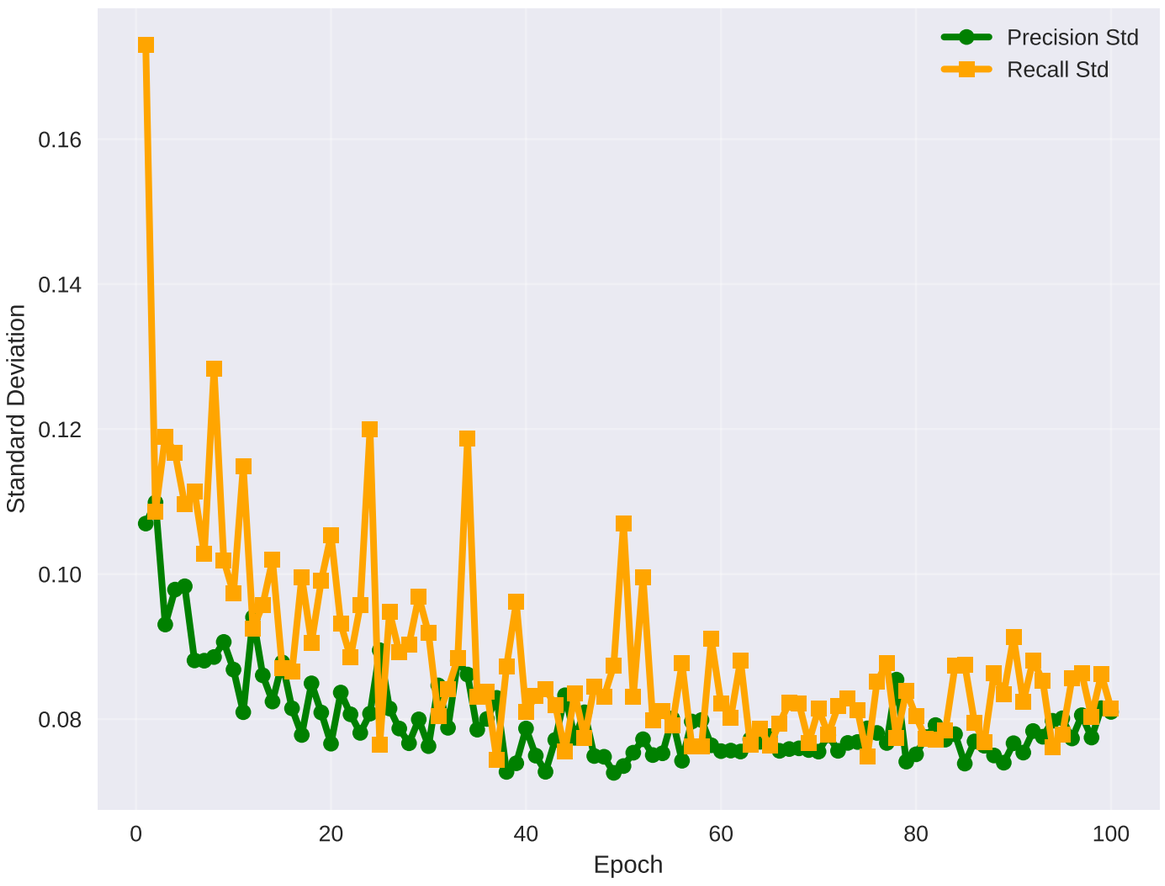


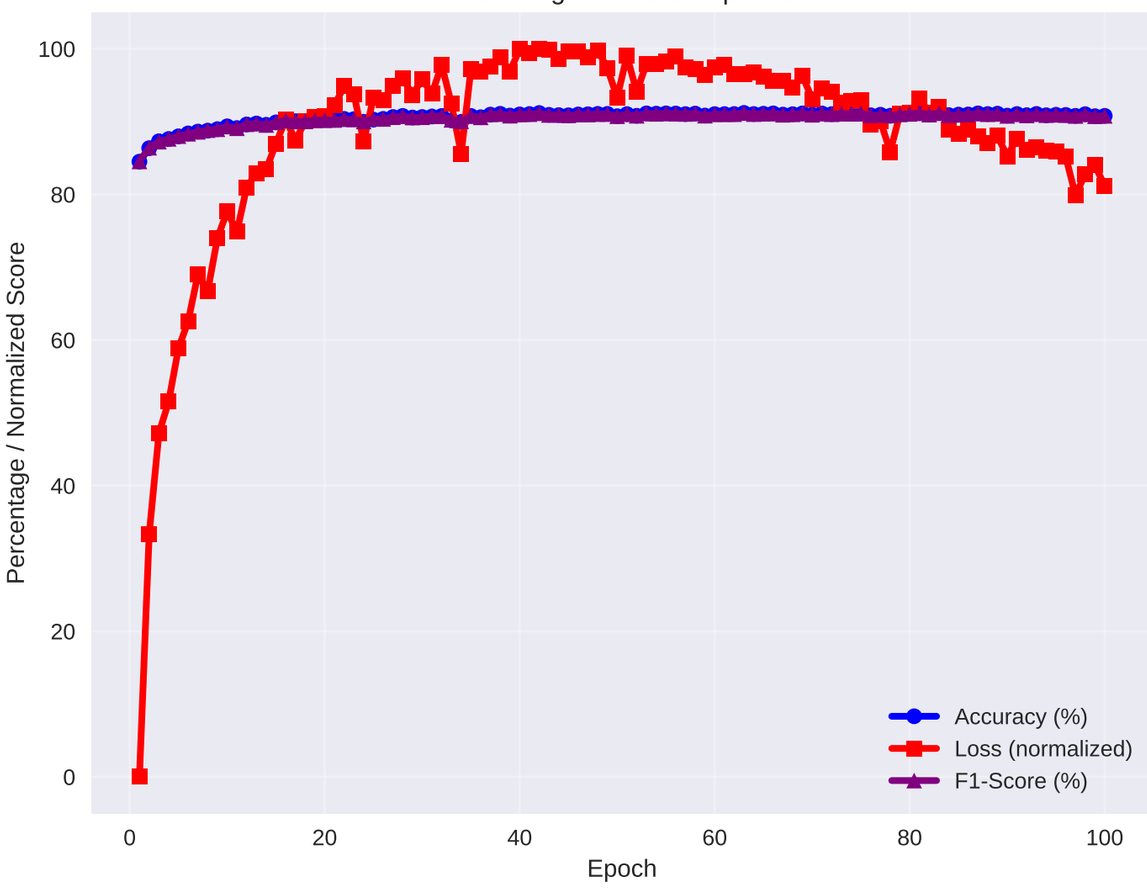


Comparative Analysis









Performance Summary Report

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FEDERATED LEARNING - PERFORMANCE SUMMARY
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FILE ANALIZZATO: 3000_results.csv
DATA ANALISI: 2025-08-31 20:59:38
EPOCHE TOTALI: 100

METRICHE FINALI:
• Accuracy: 90.79%
• Loss: 2.9165
• Average Precision: 0.908
• Average Recall: 0.908
• Average F1-Score: 0.908

EVOLUZIONE TRAINING:
• Miglioramento Accuracy: +6.29%
• Riduzione Loss: +1.4843
• Stabilità Accuracy (CV): 0.0120

PERFORMANCE PER CLASSE:
• Migliore Precision: Classe 1 (0.994)
• Peggiore Precision: Classe 6 (0.747)
• Migliore Recall: Classe 8 (0.978)
• Peggiore Recall: Classe 6 (0.729)

ANALISI DETTAGLIATA PER CLASSE:
• Classe 0: P=0.842, R=0.859, F1=0.850
• Classe 1: P=0.994, R=0.973, F1=0.983
• Classe 2: P=0.849, R=0.883, F1=0.866
• Classe 3: P=0.893, R=0.915, F1=0.904
• Classe 4: P=0.873, R=0.839, F1=0.856
• Classe 5: P=0.982, R=0.970, F1=0.976
• Classe 6: P=0.747, R=0.729, F1=0.738
• Classe 7: P=0.949, R=0.971, F1=0.960
• Classe 8: P=0.983, R=0.978, F1=0.980
• Classe 9: P=0.969, R=0.962, F1=0.965

VALUTAZIONE QUALITATIVA:
• Performance Globale: BUONA
• Stabilità Training: STABILE
• Balance Classes: SBILANCIATO

RACCOMANDAZIONI:
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